

Quantitative Trading and Price Impact

End-to-end pipeline with non-parametric impact
and multi-day carry backtest

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*Modular src/price_impact/ package | thin orchestration notebook | every backtest run writes to
saved/<name>/*

Pipeline at a glance

Seven assignment sections, eight modules, two enhancements

Modules vs assignment sections

Module	Responsibility	Section
data	top- N , σ /ADV	§2.1
impact_states	OU $\bar{I}$ (daily + multi)	§2.2/§2.3
fitting	OLS + NP + H -grid	§2.2
alpha	synthetic α	§2.4
strategy	OW/AFS/ext-OW	§2.5
backtest	Waelbroeck sim	§2.3
results	metrics + TCA + plots	§2.6
runner	BacktestConfig→disk	§2.6/§2.7

Design principles

- Single-responsibility modules; notebook is glue
- Same simulator runs **any** signed trade path
- Carry mode = one flag; downstream unchanged
- Every run dumps tables + plots to saved/<name>/

Two enhancements emphasised: **NP impact** with cross-stock γ shrinkage (§2.2) • **Multi-day carry** backtest engine (§2.3)

Solo submission – all sections by author.

Panel

- 2019 US equities, 10-s bins (monthly CSVs ~200 MB each)
- **Top-20 stocks** by full-year total $|q_t|$
- Final panel: ~11.0M bin rows, 250 trading days

Daily statistics (trailing 20-day rolling)

$$\sigma_{i,d}^2 = \overline{\text{Var}_t r_{i,d,t}}, \quad \text{ADV}_{i,d} = \overline{\sum_t |q_{i,d,t}|}$$

Used to normalise flow so impact coefficients are comparable across stocks and days.

In-sample / out-of-sample protocol

10 rolling windows:

- train on month m
- tune γ on $m+1$ (NP only)
- validate on $m+2$

for $m = 1, \dots, 10$.

```
build_panel(data_dir, year,
            top_n=20,
            lookback_days=20)
→ PanelData(bins, daily_stats,
            stocks)
```

Module data = 165 lines, deterministic, no I/O outside the bin CSVs.

Both models build the OU recursion

$$\bar{I}_{t+1} = (1 - \beta)\bar{I}_t + \tilde{q}_t, \quad \beta = \frac{\ln 2}{H_{\text{bins}}}.$$

They differ only in the kernel applied to raw flow:

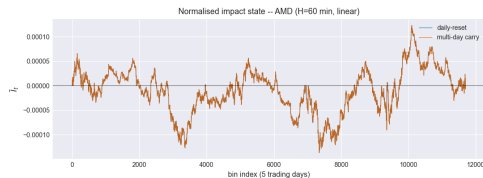
OW: $\tilde{q}_t = \sigma q_t / \text{ADV}$

AFS: $\tilde{q}_t = \sigma \text{sgn}(q_t) \sqrt{|q_t| / \text{ADV}}$

The sqrt form is chosen so the model reproduces the empirical meta-order law $\Delta P / P_0 \propto \sqrt{Q/V}$ (Almgren et al.; Bouchaud): a constant-rate parent of size Q builds equilibrium $\bar{I}^* = f(Q/T)/\beta$, whose shape inherits f . OW would predict linear-in- Q impact, which the data rejects.

Two carry modes produced side-by-side

- `I_bar_daily` – reset at each session
- `I_bar_multi` – carry with overnight decay $(1 - \beta)^{16h \cdot 6}$



At $H = 60 \text{ min}$ the overnight decay $(1 - \beta)^{5760} \approx 10^{-5}$ already wipes $\bar{I}$, so the two curves overlap on screen. The flag still matters for the backtest because **inventory** carries even when $\bar{I}$ does not.

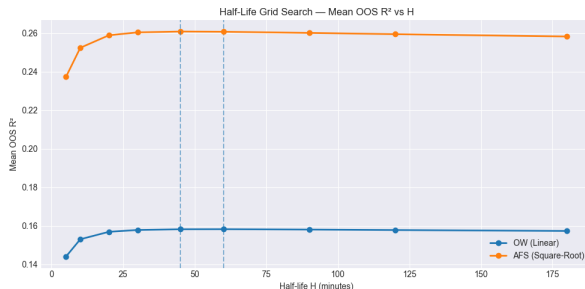
Optimisation problem. Joint $\max_{H, \alpha, \lambda} R^2_{\text{OOS}}(H, \alpha, \lambda)$. For fixed H , OLS pins $(\hat{\alpha}, \hat{\lambda})$ in closed form, so the problem reduces to a 1-D scan over H – the half-life that yields the best-fitting impact regression:

$$\max_H R^2_{\text{OOS}}(H, \hat{\alpha}(H), \hat{\lambda}(H)).$$

Grid:

$H \in \{5, 10, 20, 30, 45, 60, 90, 120, 180\}$
min.

H (min)	OW	AFS
30	0.1579	0.2604
45	0.1582	0.2609
60	0.1583	0.2608
90	0.1581	0.2602



Why the midpoint? $H^* = 52.5$ min.

- Per-model optima differ: OW peaks at 60, AFS at 45
- Surface **flat in 45–60 min**: $< 10^{-4} R^2$ cost anywhere in band
- Midpoint is model-agnostic; lets OW + AFS share bins + comparisons

Replace $y = \alpha + \lambda x$ with a binned $g(x)$:

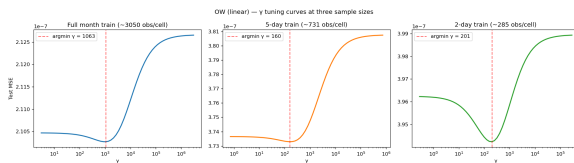
$$g_{i,b}^{\text{reg}}(\gamma) = \frac{S_{y,i,b} + \gamma \bar{g}_b}{n_{i,b} + \gamma}$$

- $B = 15$ quantile bins on train month m (pooled across stocks)
- $\bar{g}_b$ = pooled universal curve
- γ **tuned on test month $m+1$** (MSE), validated on $m+2$

Why three samples? Tuning γ on the validation month leaks. We need *train*, *tune*, *validate* – distinct months, in order.

```
rolling_nonparametric(features,
    n_bins=15, n_windows=10)
→ (summary DF, fits dict)
```

Regime-correctness stress. Headline ($n \approx 3000/\text{cell}$): tuning curves are flat $\Rightarrow$ shrinkage has nothing to fix. Stress: shrink train window 1 mo $\rightarrow$ 5 d $\rightarrow$ 2 d so per-cell n drops $10\times$.



Train	Med. n	OW gain (reg–raw)	γ^*/n
1 month	~ 3000	+0.0005	0.16
5 days	~ 720	+0.0025	0.56
2 days	~ 290	+0.0070	0.93

OOS gain grows $\sim 14\times$ as data shrinks $\sim 10\times$; shrinkage strength rises $6\times$. The estimator is **regime-correct**: trusts per-stock data when there is plenty, pools toward universal when starved.

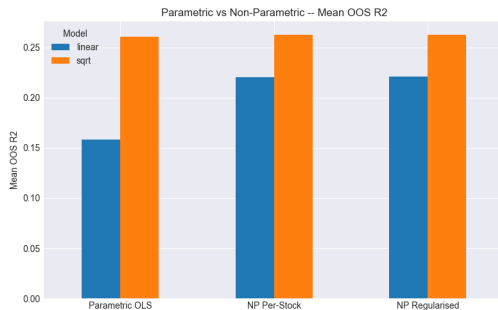
Pooled OOS R^2 (panel-wide null, same merged subset)

Model	Param.	NP Reg.	Δ
OW	0.1612	0.2206	+0.059
AFS	0.2689	0.2629	-0.006

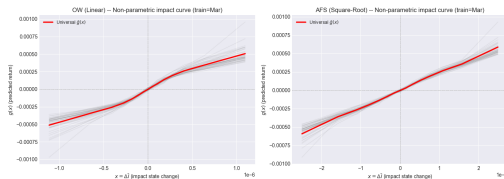
Two findings

- NP lifts OW by +5.9 pp – linearity is a binding constraint
- NP ties AFS (-0.6 pp): the 1-parameter sqrt form sits on the efficient frontier; 15 parameters buy nothing

Aggregation matters: under per-stock R^2 averaging, NP appears to beat AFS by +0.2 pp – an artefact that disappears under pooled R^2 .



OOS R^2 : parametric OLS vs NP estimators



$$P_{\text{sim}}(t) = P_0(1 + \text{cumret}(t) + g_{\text{sim}}(t) - g_{\text{ref}}(t)), \quad g(t) = \lambda \bar{I}(t) \text{ or } \lambda(t) \bar{I}(t)$$

$\bar{I}^{\text{ref}}$ uses public-tape orderFlow; $\bar{I}^{\text{sim}}$ uses the simulator's input trade path.

Generic trades. Engine accepts any signed schedule via a `trade_provider` callable:

- `make_optimal_provider(strategy_fn)` – OW/AFS
- `make_fixed_provider(trades)` – TWAP/VWAP/manual

Carry modes.

- `daily` – reset $\bar{I}$ + inventory each session (baseline)
- `multi` – persist across sessions, overnight decay $(1 - \beta)^{16h \cdot 6}$

Stock splits. Heuristic detector flags $|\Delta p|$, $|\Delta v|$ large + opposite-signed with stable notional. On 2019 top-20: **no splits detected**; engine accepts `exclude_stocks` for hard-coded exclusion.

```
run_backtest(merged, model, trade_provider,
             daily_stats, carry='daily'|'multi',
             overnight_minutes=16*60)
→ BacktestResult
```

Synthetic alpha (§2.4)

$$\alpha_t = r_t^h + y \frac{dW_t}{P_t}, \quad y = \sqrt{(\rho^{-2} - 1) \frac{\text{Var}(r^h)}{E[P^{-2}]h}}$$

Choosing $x=1$ in $\alpha = x r^h + y dW/P$ gives:

- **Unbiased:** $E[r^h | \alpha_t] = \alpha_t$
- **Target correlation:** $\text{Corr}(\alpha, r^h) = \rho$

We use $\rho = 0.05$, $h = 1$ bin (10 s); empirical ρ within $\pm 1\%$ of target.

OW closed form (§2.5)

$$X_t^* = \frac{\alpha_t \text{ADV}}{2\lambda\sigma}, \quad \kappa = \beta = \frac{\ln 2}{H_{\text{bins}}}$$

Position adjusts to X^* at speed κ ; ramped to zero over the final 30 min.

AFS variant. Same target rule per `project.ipynb`; sqrt nonlinearity enters via the simulator's $\tilde{q}$.

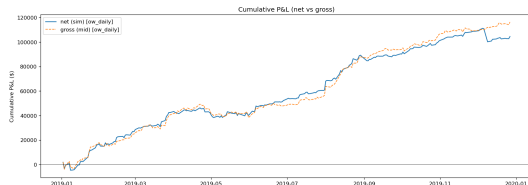
Ext-OW with $\lambda(t)$. **Placeholder.** Simulator already accepts a per-bin λ_t array; HJB-optimal trade rule is not κ -constant when λ varies – closed form pending.

Position trajectory sketch: X_t rises toward X_t^ at exponential speed κ , clips at $\pm 0.5\%$ ADV, ramps linearly to zero in the last 30 min. OW and AFS positions coincide; the cost accounting differs.*

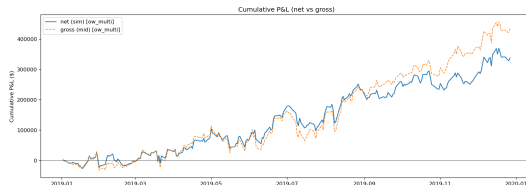
Performance – cumulative P&L

Daily-reset vs multi-day, OW model

§2.6



ow_daily: **Sharpe +4.30**, \$105k net, \$12k impact



ow_multi: Sharpe +1.61, **\$338k net**, \$98k impact

Net P&L (blue solid) marks to the Waelbroeck-distorted P_{sim} ; gross P&L (orange dashed) marks to the undistorted mid.

- **Daily-reset wins on Sharpe** – consistent recycling, tight drawdowns (−\$11k vs −\$83k)
- **Multi-day wins on dollars** – $\sim 3\times$ net P&L from accumulated inventory, but $\sim 8\times$ impact cost
- Turnover essentially identical between carry modes (\$84M) – only the *cost attribution* changes

TCA decomposition (totals)

§2.6

Net = Gross – Impact, with α rewards separated

Config	Gross P&L	Impact	Net P&L	Pred. α	Real. α	Turnover
ow_daily	116,311	11,787	104,524	44.6 M	116,311	84.0 M
ow_multi	435,399	97,813	337,586	44.6 M	435,399	84.0 M
afs_daily	119,205	56,417	62,787	44.6 M	119,205	86.6 M
afs_multi	482,557	129,999	352,558	44.6 M	482,557	86.6 M

- **Impact cost = Gross – Net** – well-defined under Waelbroeck construction
- **Realised α reward** $\sum_t \text{fwd_ret}_t \cdot X_t \cdot P_t \equiv$ Gross P&L by construction
- **Predicted α reward** $\sum_t \alpha_t \cdot q_t \cdot P_t$ – dominated by α 's noise component at $\rho = 0.05$ (only $\rho^2 = 0.25\%$ of $\text{Var}(\alpha)$ is signal); informative *across* configs, not for level
- Slippage components needing fill-price detail (arrival vs VWAP) intentionally omitted

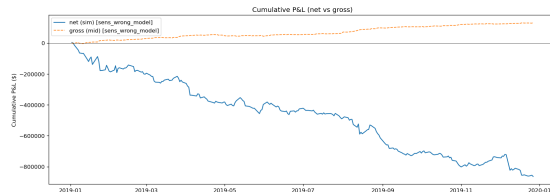
Live sensitivity sweeps (OW, daily-reset)

Scenario	Sharpe	Net P&L
Baseline	+4.30	\$105k
Wrong model (OW $\rightarrow$ AFS)	-3.48	-\$863k
Signal delayed 1 min	-0.32	-\$8k
$\rho = 0.01$	+0.45	\$11k
$\rho = 0.10$	+9.22	\$251k
$H = 30$ min	+5.91	\$205k
$H = 120$ min	+3.26	\$53k

Takeaways

- **Wrong model catastrophic** – -\$863k
- **1-min delay flips the sign** – α is short-horizon
- ρ scales Sharpe linearly:
 $\{0.01, 0.05, 0.10\} \rightarrow \{0.45, 4.30, 9.22\}$
- H is forgiving: $\pm 2\times$ stays in $[+3.3, +5.9]$

Anatomy of the worst case. Fit OW $\hat{\lambda}$ on the panel, then trade as if impact were AFS (sqrt). Year-long net P&L:



Gross alpha-capture (orange dashed) still builds steadily – the signal is real. But OW-sized positions under AFS impact pay ~\$995k in impact cost, dwarfing gross P&L. Net (blue) collapses early.

$\Rightarrow$ **Model risk dominates signal risk.**

What we learned

Re-stating the two enhancements with one bullet each

Enhancement 1 – Non-parametric impact with γ regularisation (§2.2)

Apples-to-apples pooled OOS R^2 : **OW** 0.161 $\rightarrow$ 0.221 (+5.9 pp, linearity binding); **AFS unchanged at 0.263** (square-root on the efficient frontier; NP buys nothing). γ shrinkage immaterial in the data-rich headline regime; γ^*/n rises 6 $\times$ and **OOS gain 14 $\times$** as the training window shrinks 10 $\times$ – the estimator behaves correctly under stress.

Enhancement 2 – Multi-day carry backtest (§2.3)

Carrying $\bar{I}$ and inventory across sessions makes the cost of impact *compound*: in the OW config, **impact cost \$12k $\rightarrow$ \$98k** ($\sim 8\times$), gross P&L **3 $\times$ larger**. Daily-reset wins on Sharpe (+4.30 vs +1.61); multi-day wins on absolute P&L. Engine handles both via a single flag; downstream code, TCA columns, and plotting are unchanged.

Honest limitations: overnight α (§2.4 to-expand) and extended-OW $\lambda(t)$ closed form (§2.5 to-expand) are flagged but not implemented; the simulator side of $\lambda(t)$ is already plumbed via `ImpactModel.lam_t_lookup`.

Thank you

Questions?

Code: `src/price_impact/` | Notebook: `final_pipeline.ipynb` | Runs: `saved/<name>/`